

MTH100 FINAL TERM PREPARATION

BY BRAINY SQUAD 60 MCQ'S

MCQ 1:

Question: Write the standard form of the equation of the circle with radius $2\sqrt{2}$ and center (3, 4).

Options:

- $(x + 3)^2 + (y + 4)^2 = 2$
 - $(x - 3)^2 + (y - 4)^2 = \sqrt{2}$
 - $(x - 3)^2 + (y - 4)^2 = 2$ ✓
 - $(x + 3)^2 + (y + 4)^2 = \sqrt{2}$
-

MCQ 2:

Question: The midpoint of the line segment PQ, here P(2, 3) and Q(4, 6) is:

Options:

- M(1, 3/2)
 - M(3, 4) ✓
 - M(3, 4.5)
 - M(6, 9)
-

MCQ 3:

Question: Write the standard form of the equation of the circle with radius 11 and center (0, 6).

Options:

- $x^2 + (y - 6)^2 = 121$ ✓
 - $x^2 + (y + 6)^2 = 11$
 - $(x + 6)^2 + y^2 = 121$
 - $(x - 6)^2 + y^2 = 121$
-

MCQ 4:

Question: The n th term of an AP is $a + (n-1)d$. If $a=5$, $d=3$, and $n=6$, what is the n th term?

Options:

- 18
 - 20 ✓
 - 30
 - 23
-

MCQ 5:

Question: The number of accidents in a city during 2020 is:

Options:

- **Discrete variable ✓**
 - Qualitative variable
 - Continuous variable
 - Constant
-

MCQ 6:

Question: The sum of an infinite GP with the first term $a=3$ and common ratio $r=0.5$ is:

Options:

- 3
 - 9
 - **6 ✓**
 - 4
-

MCQ 7:

Question: $\cos^2 A = \frac{1 + \cos 2A}{2}$

Options:

- $2\cos^2 A - 1$
 - $1 - \cos^2 A$
 - $2\sin A \cos A$
 - **$\cos^2 A + \sin^2 A = 1$ ✓**
-

MCQ 8:

Question: Which branch of mathematics is most closely related to trigonometry?

Options:

- **Geometry ✓**
 - Calculus
 - Statistics
 - Algebra
-

MCQ 9:

Question: $-16x+64+(y+3)^2=49$ $-16x + 64 + (y + 3)^2 = 49$ $-16x+64+(y+3)^2=49$.

Options:

- $(h, k) = (-3, 8); r = 7$
 - $(h, k) = (3, -8); r = 49$
 - **$(h, k) = (8, -3); r = 7$ ✓**
 - $(h, k) = (-8, 3); r = 49$
-

MCQ 10:

Question: Sine is positive in the:

Options:

- Third and the fourth quadrant
 - Second quadrant
 - **First and the second quadrant ✓**
 - Third quadrant
-

MCQ 11:

Question: The domain of inverse sine function and inverse cosine function is same.

Options:

- **False ✓**
 - True
-

MCQ 12:

Question: The point where the x- and y- axis cross. (0,0)

Options:

- Slope
 - **Origin ✓**
 - Y-intercept
 - X-intercept
-

MCQ 13:

Question: $\sin(\theta+2\pi)=\sin(\theta+2\pi)=\sin(\theta+2\pi)=$

Options:

- **$\sin \theta$ ✓**
 - $\cos \theta$
 - $-\cos \theta$
 - $-\sin \theta$
-

MCQ 14:

Question: What is the period of the following tangent function? $f(x)=3\tan\left(\frac{\pi}{5}\right)(5x)+4$ $f(x) = 3\tan(5x) + 4$

Options:

- $\pi/5$ ✓
- 3π
- $\pi/4$
- 3

MCQ 15:

Question: The amplitude of the function $y=9\cos x$ is...

Options:

- 9 ✓
- 3
- -9
- -3

MCQ 16:

Question: Equation of a circle centered at (0, -5) and radius $8\sqrt{8}$ is...

Options:

- $x^2+(y-5)^2=8$
- $x^2+(y+5)^2=64$
- $x^2+(y+5)^2=8$
- $x^2+(y-5)^2=64$ ✓

MCQ 17:

Question: There are 3 quadrants.

Options:

- False ✓
- True

MCQ 18:

Question: Equation of a circle centered at origin and radius “1” is...

Options:

- $x^2+y^2=1$
- $11x^2+11y^2=121$
- $x^2+y^2=11$
- $x^2+y^2=1$ ✓

MCQ 19:

Question: Equation of a circle centered at $(-5,0)$ and radius $9\sqrt{9}$ is...

Options:

- $x^2+y^2=81$ $x^2 + y^2 = 81$
 - $(x+5)^2+y^2=9$ $(x + 5)^2 + y^2 = 9$
 - $(x-5)^2+(y+5)^2=9$ $(x - 5)^2 + (y + 5)^2 = 9$
 - $(x+5)^2+(y-5)^2=9$ $(x + 5)^2 + (y - 5)^2 = 9$ ✓
-

MCQ 20:

Question: An equation that is not an identity is called a(an) _____ equation.

Options:

- Linear
 - Identical
 - **Conditional** ✓
-

MCQ 21:

Question: In how many ways can the letters of the word "MATH" be arranged?

Options:

- 12
 - 16
 - 18
 - **24** ✓
-

MCQ 22:

Question: Using the distance formula, find the distance between the two points A(4, 7), B(3, -2):

Options:

- 7
 - 19
 - **$\sqrt{82}$** ✓
 - 20
-

MCQ 23:

Question: The trigonometric functions with the property that they keep repeating themselves are called:

Options:

- Tangent functions
- Sine functions
- **Periodic functions** ✓
- Cosine functions

MCQ 24:

Question: $\sin(-\pi/4) = \sin(-\pi/4) = \sin(-\pi/4) =$

Options:

- $\sin(\pi/4) = \sin(\pi/4) = \sin(\pi/4)$
- $-\sec(\pi/4) = -\sec(\pi/4) = -\sec(\pi/4)$
- $\csc(\pi/4) = \csc(\pi/4) = \csc(\pi/4)$
- $-\sin(\pi/4) = -\sin(\pi/4) = -\sin(\pi/4)$ ☒

MCQ 25:

Question: Which of the following equations represents a tangent function with a period that is 0.252 radians?

Options:

- $f(x) = \tan(0.25x)$ ☒
- $f(x) = 0.25 \tan(x)$
- $f(x) = 4 \tan(x) - 15$
- $f(x) = 3 \tan(4x) - 15$

MCQ 26:

Question: The value of $\cos(120^\circ)$ is the same as the value of:

Options:

- $\cos 60^\circ$
- $-\cos 120^\circ$
- $-\cos 60^\circ$ ☒
- $\cos 180^\circ$

MCQ 27:

Question: The cosine function has a "period":

Options:

- 180°
- 270°
- 90°
- 360° ☒

MCQ 28:

Question: Using the distance formula, find the distance between the two points A(4, 7), B(3, -2):

Options:

- 7
- 19

- $\sqrt{82}$ ☒
- 20

MCQ 29:

Question: $\tan \theta = ?$ $\tan \theta = ?$

Options:

- $\frac{\sin \theta}{\cos \theta}$ ☒
- $\frac{\cot \theta}{\cos \theta}$
- $\frac{\cos \theta}{\sin \theta}$
- $\frac{\cot \theta}{\sin \theta}$

MCQ 30:

Question: The function $y = \sin 4x$ has a period:

Options:

- $\pi/4$
- $\pi/2$ ☒
- π
- 2π

MCQ 31:

Question: The equation of a line passing through points (1, 4) and (3, 10) is:

Options:

- $y + 10 = 2(x - 3)$
- $y - 10 = 3(x - 2)$
- $y + 4 = 2(x - 1)$
- $y - 4 = 3(x - 1)$ ☒

MCQ 32:

Question: $\tan \theta$ is positive in the _____ and the third quadrant.

Options:

- 1, 3
- 2, 4
- 1, 2
- 3, 4

MCQ 33:

Question: The amplitude of a function is the height from the rest value to the extreme value.

Options:

- True ☒
 - False
-

MCQ 34:

Question: $\cos \theta$ is negative in the second and the _____ quadrant.

Options:

- 1
 - 2
 - 3 ☒
 - 4
-

MCQ 35:

Question: Pieces of numerical and nonnumerical information are called data.

Options:

- True ☒
 - False
-

MCQ 36:

Question: The Sine function has a "period" of:

Options:

- 2π ✓
 - π
 - $\pi/2$
 - 1
-

MCQ 37:

Question: The range of $y = \cos(x)$ is:

Options:

- $[1, 1]$
 - $(-\infty, \infty)$
 - $[0, \infty)$
 - $[-1, 1]$ ✓
-

MCQ 38:

Question: $\tan(x)$ is positive in the _____ and the third quadrant.

Options:

- 1 ✓
- 2

- 3
- 4

MCQ 39:

Question: Find the midpoint of the line segment joining the points $P(1,3)$ and $Q(6,8)$.

Options:

- $(3.5, 5.5)$ ✓
- $(7, 11)$
- $(5, 5)$
- None of these

MCQ 40:

Question: The equation $x^2 + y^2 - 2x + 4y + 5 = 0$ represents:

Options:

- None of these
- A pair of straight lines
- A circle ✓
- A point

MCQ 41:

Question: The function $y = \sin 3x$ has a period of:

Options:

- 2π ✓
- π
- $\frac{\pi}{3}$
- $\frac{\pi}{6}$

MCQ 42:

Question: The variables whose calculation is done according to the height, length, and weight are categorized as:

Options:

- Continuous Variables ✓
- Discrete Variables
- Flowchart Variables
- Measuring Variables

MCQ 43:

Options:

- MCQ 44:**

Options:

- MCQ 45:**

Options:

- Correct Option: Interior ✓**

MCQ 46:

Options:

- MCQ 47:**

Options:

- 10

MCQ 48:

Question: The smallest interval for which the trigonometric function repeats itself is called its _____.

Options:

- Angle
- Amplitude
- Range
- **Period ✓**

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The smallest interval for which the trigonometric function repeats itself is called its _____.

Select the correct option

☒	period
☐	amplitude
☐	angle
☐	range

$$\cos 2A = \underline{\hspace{2cm}}.$$

Select the correct option



$$\cos^2 A + \sin^2 A$$



$$1 - \cos^2 A$$



$$2\cos^2 A - 1$$



$$2 \sin A \cos A$$

Click to Save Answer

The function $y = \sin 3x$ has period...

Select the correct option

☐	π
☐	$\frac{\pi}{2}$
☐	2π
☒	$\frac{2\pi}{3}$

Click to S

The range of cosine function is between -1 and 1 .

Select the correct option



False



True

The _____ of a trigonometric function is the height from the mean (or the rest) value of the trigonometric function to its maximum or minimum value.

Select the correct option

☐	angle
☐	period
☐	range
☒	amplitude

Click to Save Answer & Move to Next Question

Two functions f and g are said to be identically equal if _____ for every value of x for which both functions are defined.

Select the correct option

 Reload

☐	$g(x) = y$
☒	$f(x) = g(x)$
☐	$f(x) = y$
☐	$f(x) + g(x)$

Click to Save Answer & Move to Next Question

The value of $\sin 135^\circ$ is same as the value of

Select the correct option

☐	$-\sin 90^\circ$
☐	$-\sin 45^\circ$
☐	$-\sin 135^\circ$
☒	$\sin 45^\circ$

The sum formula for $\cos(A + B)$ is

Select the correct option

☒	$\cos A \cos B - \sin A \sin B$
☐	$\sin A \cos B + \cos A \sin B$
☐	$\cos A \cos B + \sin A \sin B$
☐	$\sin A \cos B - \cos A \sin B$



Click to Save

$$\sin\left(-\frac{\pi}{4}\right) = \dots\dots$$

Select the correct option



$$\operatorname{cosec}\left(\frac{\pi}{4}\right)$$



$$-\sec\left(\frac{\pi}{4}\right)$$



$$\sin\left(\frac{\pi}{4}\right)$$



$$-\sin\left(\frac{\pi}{4}\right)$$

Determine the slope of the line that passes through (2, 2) and (5, 8).

Select the correct option



2



-2



$10/7$



$1/2$

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